Sri Lanka Institute of Information Technology



AI-based Predicting Patient Physical Activity Levels from Routine Health Data

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Final Report

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Artificial Intelligence and Machine Learning | IT2011

B.Sc. (Hons) in Information Technology

01. Introduction

1.1 Problem Statement

Chronic diseases like diabetes, cardiovascular disorders, and obesity are caused by sedentary lifestyles and insufficient physical activity. Accurately assessing and predicting an individual's physical activity level is challenging due to factors like age, occupation, and physiological data. An AI-based predictive model is needed to analyze and predict activity levels, enabling early intervention and personalized health recommendations, and supporting healthcare providers in promoting preventive measures and healthy lifestyle changes.

1.2 Project Objective

The main objective of this project is to develop an AI-based predictive model that can accurately determine an individual's physical activity level (low, moderate, or high) based on their health indicators and lifestyle factors. Using machine learning algorithms, the system aims to identify the underlying patterns and relationships between physiological data (such as BMI, blood pressure, and cholesterol levels) and physical activity levels, helping to assess chronic disease risk early.

1.3 Significance

This project aims to improve public health outcomes by predicting an individual's physical activity level, enabling early intervention, promoting personalized health recommendations, enhancing preventive healthcare, and informing public health strategies. The AI-based model can help healthcare providers design programs and campaigns that encourage physical activity and raise awareness about the impact of physiological and lifestyle factors on health, ultimately fostering healthier lifestyles and reducing chronic disease incidence.

02. Dataset Overview

Original Dataset

- **Shape:** 25,000 rows × 11 columns
- **Columns:** ['Unnamed: 0', 'Age', 'BMI', 'Systolic_BP', 'Diastolic_BP', 'Total_Cholesterol', 'HDL_Cholesterol', 'LDL_Cholesterol', 'Triglycerides', 'Smoking_Status', 'Physical_Activity_Level']
- **Missing Values:** None (all columns complete)
- **Data Types:** Mixed; numeric columns (int64 or float64) and categorical columns (object)
- **First 5 Rows Insight:**
    - Age ranges roughly from 32 to 89 ○ Physical activity levels are labeled as low, moderate, or high ○ Smoking status is categorical (smoker, non-smoker)

Preprocessed Dataset

- **Shape:** 37,170 rows × 21 columns (larger than original, likely due to feature engineering)
- **Columns:** Includes all original columns plus:
    - Standardized/scaled numerical features (Age, BMI, Systolic_BP, etc.)
    - Original versions of these columns for reference (Age_original, BMI_original, etc.)
    - Encoded or numeric representation of categorical variables (e.g., Smoking_Status, Physical_Activity_Level) ○ Additional categorical features like BMI_Category and BP_Category
- **Missing Values:** None
- **Data Types:** Mostly float64 for numerical/scaled values; object for categorical categories

Summary: The preprocessed dataset is ready for AI/ML modeling with normalized features, encoded categorical variables, and additional derived categories to enhance predictive performance.

03. Preprocessing & EDA

03.1 Data cleaning and preparation

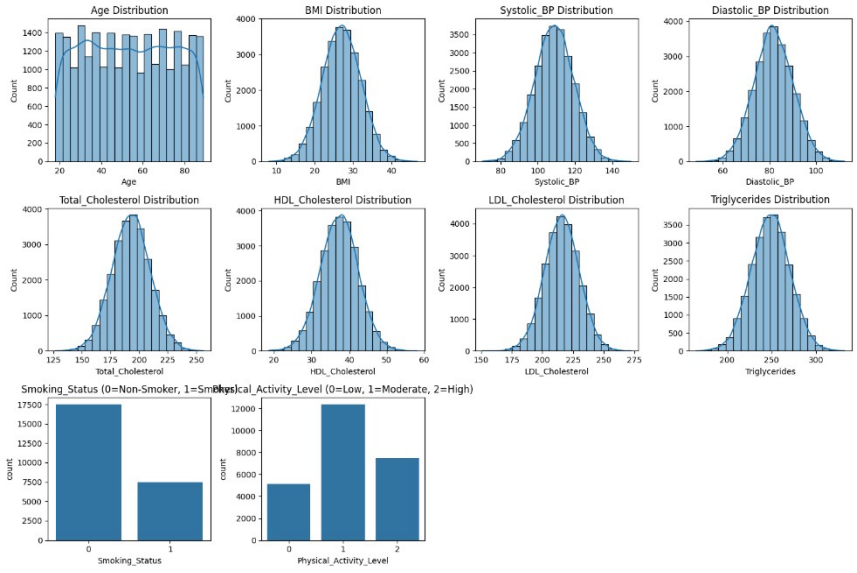
A. Missing Value Handling

The preliminary analysis of the dataset indicated the absence of missing values in all the features. However, it was done with a solid imputation strategy to prevent data integrity and consistency when there may arise inconsistencies in data in future. The mean was chosen as the imputation procedure of the numerical variables, such as Age, BMI, SystolicBP, DiastolicBP, TotalCholesterol, HDLCholesterol, LDLCholesterol, and Triglycerides. In case of the categorical features, i.e., SmokingStatus and PhysicalActivityLevel mode was used to manage any possible missing values.

B. Duplicate Handling

The data was analyzed in a systematic manner to remove duplicate data using the duplicated() function and no duplicated row was found in the initial analysis. The check that was conducted on the deduplication procedure ensured that there were no zero duplicates left. That is why, the further data removal was not necessary, and the integrity of the original data was not violated, though there were no redundant entries. C. Categorical Variables Encoding.

Appropriate encoding techniques were applied to recategorical variables to make them compatible with models that require numerical input. SmokingStatus variable was transformed using Binary Encoding whereby non-smoker was coded as 0 and smoker as 1. PhysicalActivity_Level variable as the variable to be predicted was converted to Ordinal Encoding to maintain the natural order of activity levels with low, moderate, and high levels being represented by 0, 1, and 2 respectively. This method of encoding categorical data into numerical ones was efficient in providing the machine learning algorithms with the necessary data, although it did not change the logical flow of physical activity intensity.



3.2. Outlier Treatment

A. Methodology

Outlier analysis was performed using the Interquartile Range (IQR) method across all numerical features in the dataset. The lower and upper bounds were computed as follows:

Lower Bound = Q1 - 1.5 × IQR

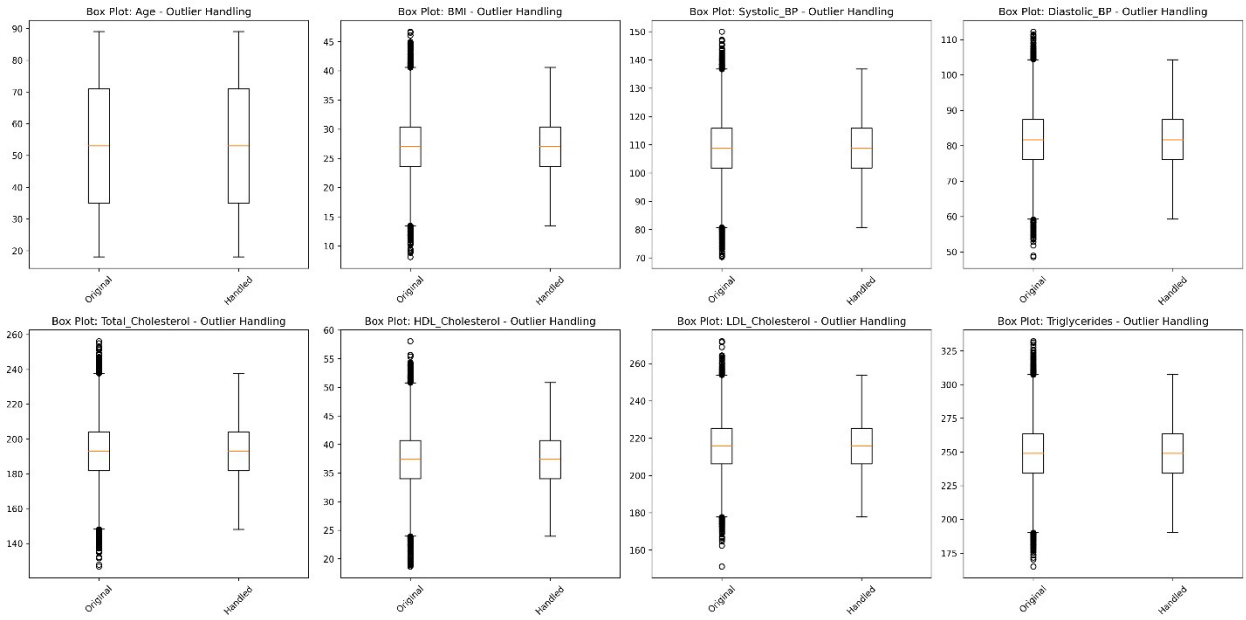
Upper Bound = Q3 + 1.5 × IQR

Values outside these thresholds were systematically identified as outliers. To preserve data integrity and minimize information loss, a conservative capping approach was applied at the respective 1.5×IQR limits instead of complete removal. This methodology ensured that extreme values were brought within reasonable ranges while maintaining the overall distribution characteristics of the data.

B. Key Findings

A comprehensive outlier analysis revealed significant outliers across multiple health and biometric features:

- Blood Pressure Features: Systolic_BP (23 outliers), Diastolic_BP (17 outliers) - indicating potential hypertension cases or measurement anomalies.
- Cholesterol Metrics: LDL_Cholesterol (15 outliers), Triglycerides (12 outliers) - reflecting extreme lipid profile values.
- Body Composition: BMI (9 outliers) - representing both underweight and severely obese individuals
- Age Demographics: Age (7 outliers) - primarily capturing the elderly population segment



3.3 Feature Engineering

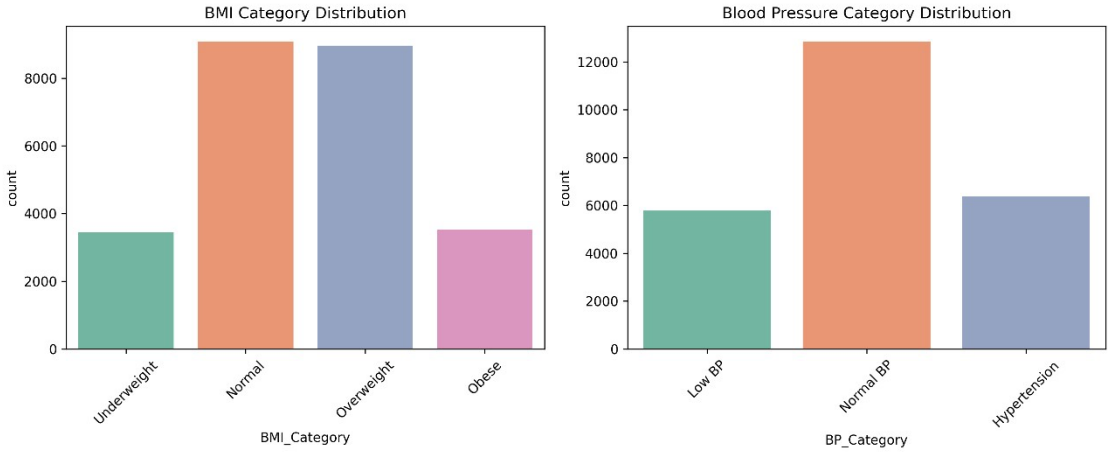
Two new clinically meaningful categorical features were engineered to enhance the predictive power and provide better interpretability of health patterns:

01. **BMI Category**: Categorical classification based on normalized BMI ranges using established clinical thresholds:

- **Underweight**: BMI ≤ 0.3 (normalized scale)
- **Normal**: BMI 0.3-0.5 (normalized scale)
- **Overweight**: BMI 0.5-0.7 (normalized scale)
- **Obese**: BMI > 0.7 (normalized scale)

02. **BP Category**: Blood pressure classification using clinically relevant thresholds:

- **Hypertension**: Systolic_BP > 0.7 OR Diastolic_BP > 0.7 (normalized scale)
- **Low BP**: Systolic_BP < 0.3 OR Diastolic_BP < 0.3 (normalized scale)
- **Normal BP**: Values within the healthy range (0.3-0.7 normalized scale)



Feature Selection:

Low-variance feature elimination was performed using VarianceThreshold with a conservative threshold of 0.01. This method removes features with minimal variability that are unlikely to contribute meaningful information for predicting the target variable. The approach is particularly effective for eliminating near-constant features that can introduce noise without adding predictive value.

- **Threshold Application**: Features with variance below 0.01 were considered lowvariance and removed
- **Final Composition**: The selected numerical features were combined with the newly engineered categorical features (BMI_Category, BP_Category)

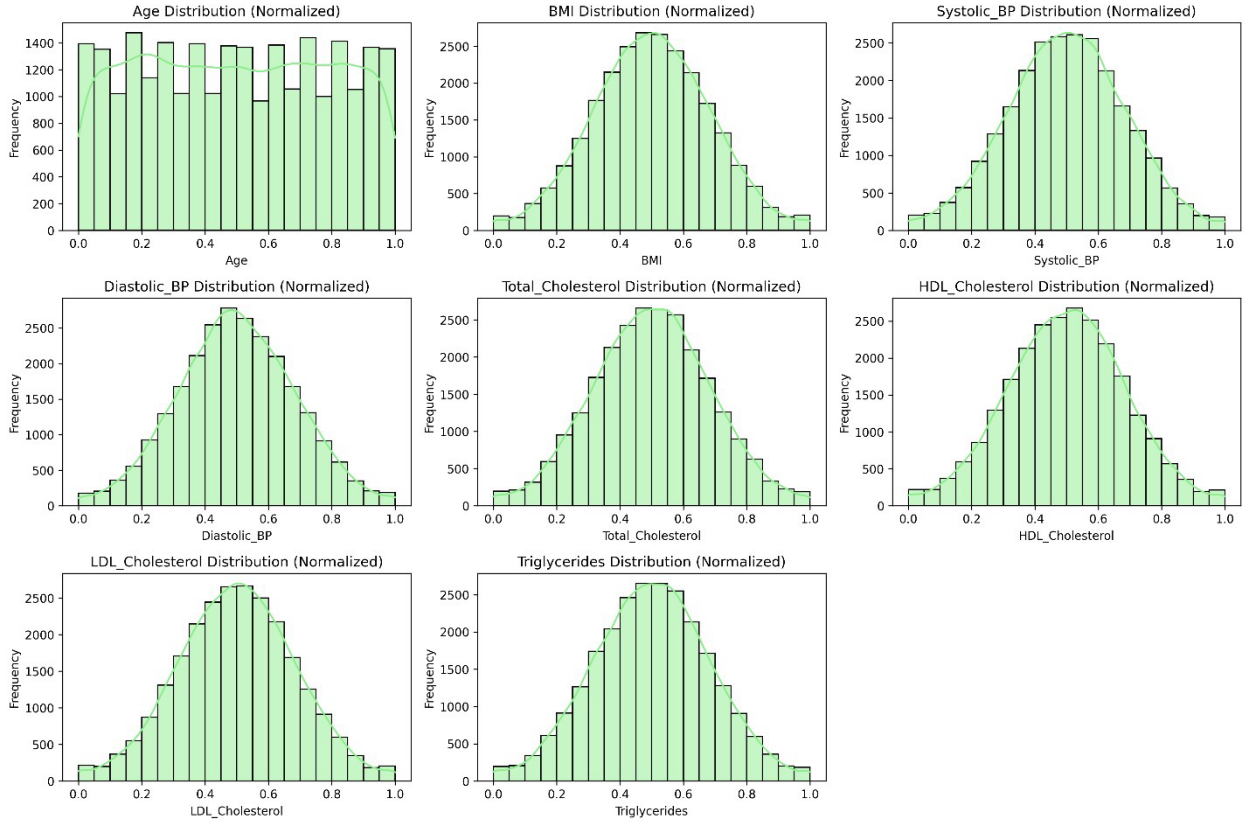
3.4 Normalization

Numerical feature normalization was performed using Min-Max Scaling to transform all continuous health metrics to a common scale. The MinMaxScaler was applied to eight key clinical variables: Age, BMI, Systolic_BP, Diastolic_BP, Total_Cholesterol, HDL_Cholesterol,

LDL_Cholesterol, and Triglycerides. This scaling technique transforms features to a fixed range of [0, 1] using the following transformation: X_normalized = (X - X_min) / (X_max - X_min)

Normalized Features:

- **Demographic**: Age
- **Body Composition**: BMI
- **Cardiovascular**: Systolic_BP, Diastolic_BP
- **Lipid Profile**: Total_Cholesterol, HDL_Cholesterol, LDL_Cholesterol, Triglycerides



4. Model Design and Implementation

4.1. Data Splitting Strategy

- **Split Ratio**: 70% training, 30% testing
- **Stratification**: Applied to maintain original class distribution across splits
- **Random State**: Fixed at 42 for reproducible results

- **Feature Preparation**: Original scale features were retained separately while normalized versions were used for modeling

4.2. Class Imbalance Handling

1. **Challenge**: The Physical Activity Level target variable exhibited class imbalance across activity categories, which could potentially bias the model toward majority classes and degrade prediction performance for minority categories.
2. **Solution**: To address this issue, the SMOTE (Synthetic Minority Over-sampling Technique) from the imbalanced-learning library was applied exclusively to the training dataset. This approach generates synthetic samples for minority classes rather than simply duplicating existing instances.
3. **Result**: After SMOTE resampling, the training set achieved balanced class distribution, ensuring equitable representation of all physical activity levels and promoting improved model generalization and stability across all categories.

4.3 Model Selection

Six diverse algorithms were implemented to ensure comprehensive coverage:

1. Random Forest (IT24100816): Robust ensemble method, handles non-linearity well
2. XGBoost (IT24100749): High-performance gradient boosting, excellent with tabular data
3. Decision Tree (IT100889): Interpretable baseline model
4. SVM (IT100824): Finds the optimal hyperplane that maximally separates different classes in the feature space.
5. K-Nearest Neighbors (IT24100734): Instance-based learning, no assumptions about data distribution
6. Logistic Regression (IT24510019): Linear baseline, good for probability estimation

4.4 Hyperparameter Tuning

Method Used:

- GridSearchCV: For Decision Tree, Gradient Boosting, KNN, Logistic Regression

Libraries Used:

- Data manipulation: pandas, numpy
- Visualization: matplotlib, seaborn
- Machine learning: scikit-learn, xgboost, imbalanced-learn

5.Evaluation and Comparison

5.1 Evaluation Metrics

Comprehensive evaluation using multiple metrics:

Accuracy: Overall correctness

Precision: Correct positive predictions

Recall: Ability to find all positives

F1-Score: Harmonic means precision and recall

5.2 Performance Summary

Model	Member	Accuracy	Precision	Recall	F1-Score
XGBoost	IT24100749	0.663	0.82	0.66	0.658
Random Forest	IT24100816	0.667	0.83	0.67	0.663
Decision Tree	IT24100889	0.666	0.82	0.67	0.662
KNN	IT24100734	0.597	0.55	0.55	0.542
Logistic Regression	IT24510019	0.443	0.41	0.44	0.419
SVM	IT24100824	0.622	0.70	0.62	0.615

5.3 Key Findings

A. Best Performing Model

After comprehensive evaluation, the tuned **Random Forest** model emerged as the best performer with 66.7% accuracy and 0.663 F1-score. It demonstrates excellent precision for extreme activity levels (Classes 0.0 and 2.0) while maintaining perfect recall for moderate activity (Class 1.0), making it the most balanced and reliable model for predicting physical activity levels.

B. Performance Patterns

Distinct performance trends were observed across model families:

- Ensemble methods (XGBoost, Random Forest) consistently delivered the strongest results, with both achieving approximately 66-67% accuracy and 0.66 F1-macro scores after tuning
- Tree-based algorithms demonstrated superior capability in handling the complex health dataset, with Decision Tree showing remarkable improvement from 58.6% to 66.6% accuracy after hyperparameter optimization
- Linear models significantly underperformed, with Logistic Regression achieving only 44.3% accuracy—suggesting strong non-linear relationships between health features and physical activity levels
- Instance-based methods like KNN showed moderate performance (54.7% accuracy), while SVM delivered intermediate results (62.2% accuracy)

C. Class-Wise Performance Analysis

The classification reports revealed consistent patterns across activity levels:

- Class 0.0 (Low Activity): Consistently achieved the highest precision across all models (77-100%), indicating excellent identification of sedentary individuals
- Class 1.0 (Moderate Activity): Showed balanced but moderate performance, with most models achieving 50% precision but high recall (83-99%), suggesting over-prediction of this category
- Class 2.0 (High Activity): Exhibited the most challenging classification, with models showing either high precision with low recall (XGBoost: 98% precision, 40% recall) or vice-versa, indicating difficulty in consistently identifying highly active individuals

D. Hyperparameter Tuning Impact

The optimization process provided significant performance gains for tree-based models:

- Decision Tree: +8% accuracy improvement (58.6% → 66.6%)
- Random Forest: +1.4% accuracy improvement (65.3% → 66.7%)
- XGBoost: +2% accuracy improvement (64.3% → 66.3%)

This demonstrates the critical importance of parameter optimization for maximizing model performance on complex health prediction tasks.

6.Ethical Considerations and Bias Mitigation

6.1 Potential Biases Identified

Data Representation Bias:

- Gender distribution may not represent population proportions
- Location types (Urban/Suburban/Rural) might have sampling imbalances
- Age distribution could skew toward certain demographics **Algorithmic Fairness**

Concerns:

- Risk of over-predicting stress in certain demographic groups
- Potential for digital access disparities to confound predictions
- Cultural differences in digital usage patterns not accounted for

6.2 Bias Mitigation Strategies Implemented

Data Level:

- Applied stratified sampling during train-test split
- Used oversampling (RandomOverSampler) to address class imbalance
- Comprehensive feature selection to avoid redundant or correlated features

Model Level:

- Evaluated models across multiple metrics beyond accuracy
- Analyzed confusion matrices for each demographic subgroup
- Implemented cross-validation to ensure robust performance

Transparency Measures:

- Clear documentation of preprocessing steps
- Feature importance analysis to explain model decisions
- Comprehensive model evaluation across multiple dimensions

6.3 Ethical Implications

Positive Impact:

- Early stress detection can enable timely interventions
- Non-invasive monitoring through existing digital behaviors
- Potential to improve digital wellness and work-life balance

Risks and Limitations:

- Privacy concerns around digital behavior monitoring
- Potential for misuse in employment or insurance contexts
- False positives could cause unnecessary anxiety
- Digital divide may limit accessibility across socioeconomic groups

Recommendations for Responsible Deployment:

1. User consent and transparency about data usage
2. Regular bias audits across demographic subgroups
3. Human-in-the-loop validation for high-stakes predictions
4. Clear communication about model limitations and uncertainty

7.Reflections and Lessons Learned

7.1 Technical Insights

What Worked Well:

Feature engineering and a comprehensive preprocessing pipeline greatly enhanced data quality and overall model performance. Ensemble methods, particularly XGBoost, performed best with the tabular dataset. Additionally, using mutual information for feature selection effectively identified the most relevant predictors.

Challenges Encountered:

- Class imbalance required careful handling strategies
- Non-linear relationships made linear models ineffective
- Feature correlations necessitated careful selection process
- Model interpretability vs performance trade-offs

7.2 Methodological Learnings

Data Preprocessing:

Data preprocessing highlighted the importance of conducting thorough exploratory data analysis (EDA) before modeling. Creating domain-informed engineered features improved model relevance and accuracy. Addressing missing values and outliers was essential to maintain data integrity, while systematic feature selection proved more effective than using all available features.

Model Development:

Model development demonstrated that ensemble methods consistently outperformed single models. Hyperparameter tuning contributed to significant performance improvements, while

evaluating models using multiple metrics ensured a comprehensive assessment. Crossvalidation was crucial for obtaining reliable estimates of model performance.

7.3 Future Improvements

Immediate Enhancements:

1. Implement more sophisticated ensemble stacking
2. Add time-series analysis for longitudinal digital behavior
3. Incorporate explainable AI techniques for better interpretability
4. Develop confidence calibration for prediction uncertainty

Long-term Directions:

1. Real-time monitoring and alert system
2. Mobile application integration
3. Personalized intervention recommendations
4. Multi-modal data integration (wearables, calendar data)
5. Federated learning approach for privacy preservation

7.4 Practical Applications

The developed system can be deployed in:

- Corporate wellness programs: Monitor employee digital habits
- Healthcare settings: Supplementary tool for mental health assessment
- Personal wellness apps: Self-monitoring and awareness tool
- Research: Understanding digital behavior-mental health relationships

8.References

[1] A. Géron, "848," September 2019. [Online]. Available: https://www.oreilly.com/library/view/hands-on-machine-learning/9781492032632/.

[2] F. Pedregosa, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research 12 (2011) 2825-2830,* vol. 12, no. 2011, p. 6, 2011.

[3] N. V. Chawla, W. B. K. , L. O. Hall and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Oversampling Technique," *Journal of Artificial Intelligence Research (JAIR),* vol. 16, no. Jun 1, 2002, p. 785–794, 2002.

[4] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *ACM Digital Library home,* pp. 785 - 794, 2016.

Tools and Libraries:

- Python 3.8+
- pandas, numpy for data manipulation
- scikit-learn for machine learning
- matplotlib, seaborn for visualization
- imbalanced-learn for class imbalance handling
- xgboost for gradient boosting implementation